

1. A hybrid rocket engine uses HTPB (SG = 0.915) as fuel and LOX (SG = 1.14) as oxidizer. The regression rate of the fuel is given by $r_f = 2.12G_{ox}^{0.8}$, with G_{ox} in kg/cm²/s and r_f in cm/s. The fuel is cast into the casing with a single cylindrical port with an inside radius of 6 cm and an outside radius of 10 cm. The length of the port is 350 cm. The LOX-HTPB burn characteristics are given in the following table.

$(\dot{m}_{ox}/\dot{m}_f)$	c^* (m/s)	γ	$\bar{M}$
2	1707	1.24	21
2.1	1746	1.20	22
2.2	1778	1.17	22
2.3	1801	1.15	23
2.4	1816	1.15	23
2.5	1823	1.14	24
2.6	1822	1.14	25
2.7	1813	1.15	26
2.8	1796	1.16	26
2.9	1771	1.17	27
3	1738	1.23	27

- a) If the oxidizer flow rate is 7 kg/s, find the mass ratio of oxidizer to fuel, the burn temperature, and γ and the molar mass of the combustion products, when the port radius is 6 cm.
- b) If the nozzle area ratio is 10, find the thrust coefficient and the specific impulse of the rocket in the vacuum when the port radius is 6 cm.

